

Apoorva Rastogi

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EDUCATION

Arizona State University

Masters of Science in Computer Science; GPA: 3.7/4

Tempe, AZ

Aug. 2021 – May 2023

Shiv Nadar University

Bachelor of Technology, Electronics & Communication Engineering; GPA: 8.2/10

Greater Noida, India

July 2016 – June 2020

- Awarded Position in Dean's List for Academic Excellence at Shiv Nadar University, Greater Noida (2017)

TECHNICAL SKILLS

Programming: Python, C/C++, SQL (Postgres), Data Structures, Algorithms, MIPS(Assembly Language), MATLAB

Developer Tools: Git, Docker, Visual Studio Code, PyCharm, IntelliJ, Eclipse, Postman, Jupyter, Mulesoft, Github, AWS Workspace, Apache Spark, Salesforce Developer Platform, Node.Js, MongoDB

Libraries: NumPy, Pandas, Matplotlib, Keras, Scikit-learn, Tensorflow, OpenCV, Seaborn, GPT-3

Course Tools and Basics: Canvas, GradeScope, Zybooks, Powerpoint, Excel, Sharepoint

EXPERIENCE

Automation Engineer - RPA & AI

XpertDox - AI Medical Coding

May 2024 – Present

Arizona, US

- Developed automation bots in Python, ensuring efficient real-time data extraction and processing with MongoDB for data storage and aggregation. Onboarded and trained new team members on legacy systems.
- Spearheaded the end-to-end ML lifecycle for a v1.0 product, taking full ownership of data acquisition and "clawing" to transition models from ingestion to full-scale production.
- Architected automated workflows for data cleaning and feature engineering, significantly increasing the reliability and accuracy of code prediction outputs.
- Optimized model performance by evaluating diverse algorithms, including Decision Trees and XGBoost, to establish rigorous benchmarks for correctness and production readiness.

Volunteer - Software Engineer

Owting

Sep. 2023 – Present

Remote, US

- Developed REST API's In NestJS and MongoDB to store data from the app interaction in the database.
- Formulated and implemented schemas and validation rules, coupled with adeptly designed aggregation pipelines and database design for optimizing data extraction across diverse tables in the MongoDB database.

Graduate/Teaching Assistant - Arizona State University

Arizona State University - CSE 230 Computer Org. and AL Programming

Jan. 2022 – May 2023

Tempe, AZ

- Actively engaged with undergraduate students as a Graduate/Teaching Assistant, fostering critical thinking skills and explaining fundamental computer science concepts. Offered personalized support through one-on-one and group doubt sessions, encouraging active participation and inquiry-based learning.
- Collaborated with fellow assistants and educators to suggest improvements to the course structure, ensuring alignment with students' interests and educational objectives. Advocated for changes aimed at enhancing engagement and learning outcomes, reflecting a commitment to continuous improvement.
- Recognized and addressed the diverse learning needs of students, providing tailored mentorship and support to help them overcome challenges and excel academically. Delivered constructive feedback on assignments and quizzes, guiding students in understanding their mistakes and offering resources for improvement.
- Led weekly office hours to explain complex concepts, resolve code-related issues, and address students' inquiries regarding assignments. Created a supportive environment conducive to learning, where students felt confident in seeking clarification and assistance.
- Empowered students to develop their problem-solving abilities and debugging skills, fostering confidence in their academic pursuits. Offered guidance and encouragement to inspire curiosity and facilitate deeper understanding of course material, contributing to students' academic growth and personal development.

RESEARCH EXPERIENCE AND PROJECTS

Game Theory Project - Stackelberg Security Games

Jan. 2023 – May 2023

MATLAB, Game Theory, (MILP) Algorithms

Arizona State University

- Conducted research on Stackelberg Security Games (SSGs) and adversary modeling techniques for handling bounded rationality in security applications.
- Implemented and analyzed the performance of existing models, such as BRQR (based on Quantal Response), MATCH and SUQR (Subjective Utility Quantal Response), using MATLAB for mixed-integer programming to model human adversary decision-making in SSGs.
- Utilized game theory concepts, such as leader-follower dynamics, resource allocation strategies, and adversary behavior modeling, to tackle security-related optimization problems.

NLP - Numerical Reasoning Using NLP and AI

Sep. 2022 – Dec. 2022

Python, GPT-3, LLM's, Prompting, Huggingface

Arizona State University

- Conducted a comprehensive analysis of GPT-3, NT-5, bert-base-uncased-pf-drop, and roberta-base-pf-drop models on the NumGLUE multi-task benchmark for evaluating arithmetic reasoning abilities.
- Explored zero-shot and few-shot prompting techniques to enhance the performance of GPT-3 on the NumGLUE dataset, while also experimenting with novel prompting techniques.
- Evaluated the performance of NT-5, bert-base-uncased-pf-drop, and roberta-base-pf-drop models, fine-tuned on the DROP dataset, to assess their suitability for solving tasks in the NumGLUE dataset.

Statistical Learning Theory - Recurrent Neural Networks

April 2022 – May 2022

Machine Learning, Gradient Descent

Arizona State University

- Conducted an in-depth review of two research papers that explored the use of Recurrent Neural Networks (RNNs) as optimization algorithms for supervised learning, deviating from their role as a supervised learning algorithm
- Analyzed the novel approach of employing RNNs for supervised learning tasks, replacing the conventional gradient descent method, with the aim of improving the learning process and achieving better performance
- Gained insights into cutting-edge research in the field of machine learning, expanding knowledge on advanced concepts, black-box optimization, and the innovative use of neural networks for algorithm optimization

Machine Learning - Gesture Recognition CNN

March 2022 – April 2022

Python, Tensorflow, Matplotlib, Neural Networks

Arizona State University

- Implemented three CNN models for hand gesture recognition using a dataset with 5,243 images and ten classes of hand gestures. Implemented using Python libraries such as TensorFlow, Scikit-learn, pandas etc.
- Conducted an extensive comparative analysis of three CNN models (AlexNet, ResNet50, MobileNet) for Gesture Recognition across various metrics, including training loss, training accuracy, and training time
- Determined that MobileNet exhibited the most consistent performance with training accuracy of 99% and training time of 2044 secs which is the least among three, making it the preferred model for the application.
- Studied and implemented predictive analytics and other machine learning techniques in Python including hierarchical Bayesian regression and Ridge regression along with clustering algorithms during course work.

Diabetes Management System Development - Data Mining

Mar. 2022 – April 2022

Python, Data Analysis

Arizona State University

- Development of a data mining project aimed at creating an effective Diabetes Management System - Employed machine learning methodologies to extract pertinent features, train models, and conduct thorough evaluations to accurately identify categorical instances, demonstrating strong problem-solving skills and data analysis.
- Studied and implemented predictive analytics and other machine learning techniques in Python including hierarchical Bayesian regression and Ridge regression along with clustering algorithms during course work.